

ActInf OrgStream #002.1 ~ Eugene Leventhal

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to organization and today should be a good one because we are here with eugene who's going to be talking to us about decentralization and smart contracts and all kinds of other things we also have this interactive whiteboard slide deck here so please write any comments in the chat and uh as soon as eugene is back i'm sure he'll begin his initial presentation but welcome everybody looking forward to seeing your comments and welcome back your re-entry please provide a introduction and some context and really looking forward to this of course i got so excited to begin with that i closed out the wrong tab so it's always a good beginning but thank you for for hosting me daniel excited to be here with the actonflab uh and to get to talk a little bit about the centralization and yeah kind of see where the conversation goes so definitely really excited to hear what folks are thinking about and so please at any point if this isn't going kind of deep enough uh on any specific term you want more clarity on anything please please feel free to uh mention it in the chat we will definitely be happy to to bring it up in conversation so just to begin i'm eugene leventhal i'm currently the head of operations at the smart contract research forum we're a grant funded organization in the smart contract space or in the web 3 space overall excuse me and our focus is very much on helping to advance research around web 3. the starting point for that is just connecting folks from industry and academia and we're exploring a lot of various questions about what is public good infrastructure for research in web 3 but also for science more broadly look like so uh very very happy to talk about smart contract research forum or scarf as we colloquially call it uh ad nauseam but yeah for for today the focus is on decentralization itself so i want to take it there but the thing i wanted to kind of start off with and and uh mention as the genesis of this conversation is the fact that the space of web 3 even the term web 3 right it's one i i came across this uh github repo of just all these articles that where it's very well known technologists uh explaining why web 3 is all a bunch of baloney and none of this makes sense and uh in all fairness there is a lot of enthusiasm in this space that leads to there probably being more focus on areas that um you know uh might have rockier grounds than others but the goal of this session is really to delve into a term that if you've been part of anything

blockchain crypto web 3 in any kind of way you've definitely heard the term decentralization thrown around because the whole dichotomy and i don't know if it's even fair to call it a dichotomy because that oversimplifies the idea but at least this idea of a spectrum between they're centralized and somewhere along the line there's decentralized um what does that actually mean to be decentralized and for me another thing which i always love thinking not in binaries but sort of in spectra is decentralization an endpoint or is it itself only a certain check mark on a much wider spectra where maybe centralization to full peer-to-peer distribution is the actual ends of it where everything goes through one party or goes through everyone in the system and there's sort of a you know gradients in between but i think right we can delve right into more of the technical definition of the architecture the history of the term of decentralization or something like that but to zoom out i kind of like thinking about decentralization from at least two high-level buckets one is who are we talking about and the other is what are we talking about the who pertains to are we talking about owners of a system or users of a system and that was users of the system so those vary greatly right because if we're talking about ownership that's where things like co-ops the whole cooperative ownership movement i mean that has had millennia behind it right that's where that pertains but there's also a there could be a scenario where users use a system that on an architectural perspective is decentralized in terms of there not being a single point of centralization i know i still haven't defined it yet that's intentional um but the for the user right they could have no ownership in the state so it's not as though we mean decentralized decentralization itself does not mean cooperatively owned per se right so let's suss apart who is actually using it and who is owning the thing that we're talking about and then when it comes to actually thinking about the what gets decentralized right i think that's where it's important to think on the one hand there's the infrastructure layer and more the application layer and alongside that it's sort of what is the actual domain of whatever we're talking about right because infrastructure or application is more of uh the the the uh the dimension of kind of uh architecture so to say and the actual domain can be in terms of knowledge right our databases are they uh with known knowledge is that easily accessible right you one can argue that uh libraries right the whole goal of or ego of libraries was to make information more accessible it's taking it away from a single place and opening it up to have more people being able to interact with it and that can relate to knowledge to decision making or communications to actual resources and so right i'm kind of dancing around a few of these concepts around decentralization but again what am i actually talking about right so for me it always comes down to if we're talking about decentralization it

needs to be in the context of a system right decentralization is not just something that appears or disappears right it's only in the context of systems and when we think of those systems again are we we can be thinking about and let's just think of this from from say the what perspective right infrastructure apps what are the actual domain of those infrastructure apps when we talk about those things centralization or decentralization can relate to power dynamics or who actually gets to make the decision of how all of these things are run at the end of the day i know in the web 3 space there's been a lot of exit to dows you know just exiting uh to community is a more broad term that i i believe came from nathan schneider and some other folks working on now the project exit to community but the whole idea of how can we actually let some of the decision making uh and uh knowledge creation in our community be both owned by and used by the same set of folks and and kind of keep expanding that set and i think that's a lovely concept but a lot of the time decentralization at least from some of the conversations around web3 that i hear they start conflating a lot of things and which is why i like breaking it down to are we actually talking about the owners and users and are we talking to the infrastructure app player or and what is the actual domain of those things and once you map that out then you can start being a little more nuanced in trying to break out what exactly it is we want to explore the idea of decentralizing because the decentralization and power go hand in hand but again you need to be more specific in your terms and thinking about which part of a system you actually want to expand that scope of power around otherwise it can be very tough to build the necessary requirements of culture and community around this because again if you are trying to truly decentralize something that means you're going off to let more people have power in the system and more people it just equals more messiness right and that's not a judgment call i i think that that can at least for anyone who's genuinely you know try to make a decision with one other people or with a hundred other people and let me know the scenario where it's easier with the hundred than with the one uh and i'm sure you can always find an exception right but as a rule of thumb uh that does seem to hold and so that's kind of one thing i wanted to to start with in terms of decentralization overall and this kind of diffusion of power and again being particular with the terms of where in the system are we thinking of power and for whom does that power apply or apply to um and now that we've gotten at least some kind of a baseline of decentralization now let's jump into what do i actually mean when i say something like decentralizing science because if you've been following some uh you know random or not random but then following certain niches of crypto twitter which i recognize crypto twitters already in niche so niche among

niches but you know you might have seen the terms *decentralizing science* or *decentralized science* so we're diffusing the power around the creation or the propagation right around what of science science is itself a very uh high level term that can mean a lot of different things to a lot of people so i like to think of right if we're breaking it down into the highest level but separate parts there's the actual research side so there's someone who has an existing base of knowledge who has a theory about something that is not captured in that base of knowledge and wants to run some kind of tests to appear to somehow amend or append the existing base of knowledge with new information so that's what i see as the actual research function then there's the review function some people looked into a thing and want to claim that it is true or not how do we double check that right so the the quality assurance layer over research and then there's the publication of research which should not be conflated with media or publications overall because research publication as it stands today and i can't speak this confidently globally but at least in the states research publication and usable kind of interactable publication are two very different things sort of you know the springers the acms the ieees their goal is not to get 10 million hits in academic paper you know they're working within a very specific system of producing academic research that is connected to the review to the existing structures of research and review when it comes to just pure media there it becomes a lot of the time how do you get it to as many people as possible which means how do you sometimes lose the depth of the point to make it easier to digest in a short amount of time which is it inherently contradictory to research which is let's go as deep as possible on as narrow a set of things as we can to actually learn something or add to some kind of base of knowledge around that more narrow deep set of things so in my sort of pipeline of high level pipeline of science it's there's research which gets reviewed and then it gets published that inherently does not capture everything out of uh coming out of academia or inherently coming out of any brilliant minds around the planet who have good ideas that you know should be explored and there's the question of who has access to be doing research in the first place right in the states today if you're not at a top 10 top 50 at least research institution you're not in a top corporation that actually has an r d department that's meaningfully funded which is not the case at all corporations and you're not at a government agency dedicated to research you're probably gonna have a hard time having a seat at the table unless you find a way to break into one of those areas right so we can start thinking of well do we mean decentralizing research means

everyone should get to do research maybe i'm not here to sort of give a yay or an a but it's starting to think of well what do we mean about the power there is it who actually gets access to be a researcher is it who gets to have a call and be part of the decision of what gets funded is it something about the actual interaction with the research information right those are all different aspects of what could we be talking about when we're talking about ad or changing around the power dynamics around research itself but again that's just the first step because then we just got to jump to review even if we've ironed all that out we have a clear sense of what it means to decentralize research well who is actually going to be responsible for quality assurance and i think that you know there's no shortage of information on some of the current challenges in peer review i don't want to distract myself now but i can pull together a link later when i have a moment a professor from carnegie mellon nihar shah he and another professor from cmu zach lipton did a great joint presentation that i i know that their slides are publicly available that i'll have linked to they did a great outline of all of the known research studies delving into the current problems with uh peer review and i i don't want to be misquoting but i think it's somewhere in the 70 to 100 range of academic studies just exploring very particular problems in peer review right so this is not as though i'm someone from outside academia raising a red flag that doesn't exist right talk to a tenured faculty member and they will probably be the first one to tell you all the problems with the peer review structure of today um so one of the challenges is if you ask the people who are doing the research to be their own reviewers right there's a clear conflict of interest there if you tell me to produce something and review it there's reason to believe that i will have the incentive to be light on the review and just push whatever i want to produce so the separation of those two functions is very important but the question of you know should reviewers be anonymous or not should reviewers be compensated or not again should everyone be able to be a reviewer should everyone have a seat at the table of being to review or do you need a bare minimum set of qualifications to even you know be allowed to be a reviewer in whatever a future system of review might be and then finally there's the aspect of the actual publication of you've done the research you've gotten it reviewed now it has to go somewhere and at least the current norm a lot of the time in the top research institution is via a journal or conference or some kind of very high level or very uh esteemed publication right you might have heard the the motto so to say publish or perish which i think is a good encapsulation of the incentive problems around academia for research faculty where a lot of the time you know if you're in the top kind of one percent of institutions you might be focused on more headline catching uh research because you might

actually be the pipeline where you have the relationships with the largest media organizations and they come to you for some kind of fancy you know new breaking research on people and the brain and this is what this means for how all of us you know the kind of stuff mainstream media likes to oversimplify when it comes to uh talking about research publicly uh on the other side of it there's also uh or sort of excuse me with the aspect of publisher parish that's just in the top one percent if you go into more of the uh the kind of the global majority of research institutions the game that's being played is different where it's a volume play where your ability to progress as a tenured faculty member has pressures on it to see how many things you've actually published that hit some kind of reputable journal and right those are all incentives that make it tough for people to just focus on the purity of the act of review and there's a whole separate question right does the purity of the act matter as long as we can figure figure out a system and function so that we minimize the amount of questionable science that gets put out right what is the the bad side of review that uh research happens that comes to faulty conclusions no one captures it and someone tries building on it right and that could actually lead to some kind of problems if we think of the hard sciences you know imagine a medical device going out uh built on a fundamental research premise that is false right like people's lives could actually be at risk in those kind of scenarios so this isn't just some like people playing with d5 money online kinds of questions right this is science more broadly so but again that whole cycle of research review and publish still doesn't actually get to how do we get this research out of the ivory walls of universities into the hands of people who need to build it how do we actually make this relevant to your average person anywhere on the planet who might wonder what is the cutting edge of uh i don't know quantum computing have to do with me if i don't even know how to type a line of code or explain how an email gets from my phone to my computer or to anything else nor do i care right i'm busy with my own life and i have my own priorities and problems why should i care about any of this in the first place right so there's already there's the full research apparatus that we talked about and a lot of the institutions that do science reporting or any kind of well let's actually get this contextualized and larger narratives right what are the actual incentives aligning them with the researchers are them with the public so there's a number of things in all of these right all of these separate nodes of the large system that is science that themselves have a multitude of different areas where we can explore what decentralization might mean and how that might impact the people in the system uh or how that might impact people who don't care about the system at all but just want to have uh you know less suffering more prosperity in their life and

how does it actually go from theoretical science to genuinely improving people's lives and the last bit i'll kind of touch on before bringing this to be the actual conversation it's intended to be um is more on the behavioral side because i know when i first got interested in blockchain endows in the 2016 timeline it felt very much that there was just so much excitement that oh we could be at the precipice of making the thing that solves all of humanity's problems right and if you've studied the history of technology or of any kind of advances that's a sentiment that is not new a lot of people had a lot of cutting edge advances at their point in time thought that oh this could be the thing that actually you know ends uh world hunger or poverty or whatever right that it could actually have these mass global scale positive effects but especially when it comes to something like decentralization i feel as though one of the questions that i'm hoping we'll get into this through conversation right but a decentralized world right if we have this kind of world where power gets distributed from a very limited set of players to a much wider base of individuals there's just more uncertainty there right because like it or not right and again look at political governance in the full history there democracies are messier in the sense of decision making because more people have to come together but there's a lot of benefits of that messiness i am personally a huge fan of that messiness because i think us working through messiness is that it's it's the most human thing we can be doing right we are a social species and us working through the messiness of coordination is part of the exciting thing is that we get to build higher order ideas uh as a species like these laptops we're talking on or any of the other great advances that we now have um and so i think at the core of decentralization for me is an increase in uncertainty and i think that that is something that we don't like talking about a lot in the space or at least i have not heard many open discussions about it because it runs counter to human nature and desire right and like let's look at that at a few different dimensions right let's just think of us as a pure biological creature let's take away all of our socio social constructs all of our socio-cultural constructs we are just a biological mass right i am personally of the camp i am not a neuroscientist but in my limited knowledge and following some neuroscientists uh i i am fully in the camp um of the lisa feldman barrett's and the many others who are now part of this growing school yeah our brains are prediction machines that's literally what they evolve to do is to be much better at predicting what's about to happen and so right what does prediction mean it means that we like to model things out right and there's a lex friedman has a great series of interviews with lisa feldman barrett and yoshobak and a few others that i'd be happy to link to that i just love those three to four hour discussions jumping around neuroscience and consciousness and

everything else that you know that makes sense to me that there's these sort of there's cortical columns that are focused on certain activities uh there are sort of columns of neurons being bunched together or really of neural nets being bunched together to sort of execute certain functions as you kind of get more and more of these together you need consolidation functions right and we have models at all different kinds of levels ranging from uh very narrowly focused ones to like hey what do i want to do with my life tomorrow or a year from now right like those are very complicated modeling exercises but our brains are constantly trying to model and modeling means you're trying to map on you're trying to limit uncertainty and create an actual projection into the future of what might happen and why does centralization show up in a lot of socio-cultural environments because there's efficiency decentralization right if one person gets to make all the calls assuming they are good at making calls and they're a good person with or that they're shooting towards a beneficial end right we will all be able to move more quickly with that one as opposed to 10 000 trying to make that decision so a lot of this natural evolution has pushed us to want to minimize uncertainty and yet we are inherently trying to introduce a cultural element that actually increases uncertainty which for me again is very just counter our biology and i think that right we don't just have to think on that level of our brain as prediction machines and along those lines right how many how easy is it right at the start of my day and i'll just use the personal example as an abstraction right at the start of my day especially as someone with chronic sleep issues and whatever i start every day at the center of my universe right and i have to be intentional about reminding myself that that voice inside my head that's yapping as soon as my eyes open up that is not the entirety of me that's actually one of the nodes that constitute the entire being and system that is eugene but at the end of the day that's the thing that i hear first and foremost and it's very easy for me to just get caught up in the emotionality of whatever that voice is yammering on about and to pretend that i need to be going with it wherever it's taking me where i have the ability through meditation mindfulness through all different kinds of practices to be like time out let's go ahead and quiet that voice and let's listen to other modes let's listen to other nodes in the system of eugene and right for me decentralization is a psychological concept too if we think of ourselves as a system do i just give credence to the centralized view of the voice of consciousness or do i actually try to have a more decentralized view of my being where i'm listening to every single node that is part of my constitution right before i started meditating i never actually thought about how does my back feel against the chair that i'm sitting in as i'm right like these little elements of focusing attention and being intentional i think those two

words of intention and attention are so important for where behavioral level change begins and what i don't see enough of in this space or in a lot of spaces is the bridge between behavioral level change and high level philosophy so that's really why i wanted to have this discussion today is because i i'm happy to espouse tom blue in the face of my my views on why certain decentralization is great and good for humanity and all this other stuff but at the end of the day you can have the best philosophical theory in the history of humanity and if you don't actually know how to get a single person to start taking a step towards it is it really useful so for me i love jumping between the two between what can i do to be a better collective citizen of the groups and systems that i'm in and jump to what does that mean to inform my philosophy of life and what is the link between those two and how do i get a feedback loop going and not just punting off to philosophizing and abstracting which is much easier than the the lower level of understanding myself emotions and all of the internal world side so that's kind of that might have gone on a little longer than i intended but that's my where i'd like to kind of kick us off okay great thank you for the introduction and opening set of ideas a lot of places to jump in there did you ever get to what decentralization is let's return to maybe that and then yeah sure so yeah it's been a few other places absolutely yeah so uh i like to think of decentralization right the overly simple view of it is it's it's a diffusion of power in a system right and that's where i came back to power uh for whom over whom and of what and the whom and the water very important there but effectively it's taking power from a single source and and sharing it out most simple version uh think of a you know a kingdom when there's literally a single king or queen right they get power over everything a democracy uh such as the uh you know let's say american democracy we have three different branches we have a bunch of people in each of the branches right that that's trying to take a step towards decentralizing power away from a single person to a larger set and then you can argue that like truly communal living or certain versions of anarchic living where there is no it's kind of a leaderless true organization where everyone has equal power that might be on sort of the other end of the spectrum of sort of true everyone is on i don't know if everyone is on the same level but like this peer-to-peer system on the one end and more of a truly centralized everything comes out of one node uh on the other end so i wrote down does that change the nature of the system if we have one set of rules or one's state that we're in or an area of state space for some system features and one person has all of the tokens and then the system changes so that the distribution the non-equilibrium steady state like we might say an active inference is at a new normal it's at a new stationarity or meta stationarity it's kind of like isn't that a different system even implying that power can

be spread out it's a non-fungible asset and it's an irreversible asset time dependent one too these are just like some of the as we open the conversation i think today but also on a broader time scale there are the known unknowns and then there's the unknown unknowns and we're going to bump up against all of them yeah and i really like the the i've never used this term exactly but i really like that you put it together the non-fungibility of power because i think that's an interesting and important one to recognize because right being aware of that especially for people who are actively designing a system and who are the first ones building it you were inherently going to have way too much power in that system whether or not you want that power right i think uh let's say say vitalik buterin and ethereum is a great example right he doesn't want to i don't right i don't know him personally i don't want to speak on his behalf but i have a lot of reason to believe that he is not looking to be the authoritarian over ethereum right that is my genuine view of that based on all the reading and following of the space that i've done but vitalik says something and everyone in the space listens and that might have impacts on ethereum price right so like there are these relationships whether or not you like it but at the same time there's also as a person designing these systems you have to be willing to recognize that in order to gain efficiency at a per at a certain point i actually might need to centralize the power under me but design the system in a way so that i cannot maintain it over time and how to give away power and make sure that you don't have the ability to usurp it again is very interesting from a system design perspective and even if you came into this discussion thinking like blockchain's a joke with three is a joke sure like it's an open question how much usefulness is going to come out of this on a hundred to thousand year timeline right i am open uh and understanding of uh the lack of clarity of the future but i just think the space at least daos are worth paying attention to because i don't see anywhere else where there's so much open experimentation in that direction of supposedly saying we're here to defuse power and then watching the actual social dynamics play and system uh structuring around it so that that's one of the reasons of cause of light just go check out some doubts it's fascinating social experiments if nothing else okay a few thoughts one thing is i lived in cooperatives for many years and so it's just so interesting to see some of those traditions failure modes patterns translated and then i know it's a big theme in web 3 where the first steps involve translating like what has already been done okay oh you've worked in an office before well now you're in a virtual office and then the future steps can be elaborations that take it into a different direction or use something that couldn't have been engineered in the physical world so we're beginning to form the motifs and the grammar for describing organizational patterns

design patterns and atlases of risk for organizations and before there was money attached to numbers it was probably hard to do counterfactuals and long-term tracking of money quantitatively and if your decision-making is um based upon a secret handshake well then it might be so esoteric that it has never been investigated doesn't mean it doesn't work it's just that now we're implementing computational governance hashtag medigov and implementing programmable money and involved in totally surveillable communications oh just vote on this poll well now you have the time and did someone vote and unvote and get into so many other pieces that weren't there in the paper ballot and that wasn't there until that was there so it's a whole new set of observations that are coming in about systems that we had already been engaged in but then where you get that second level cybernetics and complexity is when we're modifying the system with new affordances new ways we can modify and then that's feeding back into like very rapid change but by no means even maximal yeah and so what you mentioned earlier on uh with the first question or with the whole idea of well once you start decentralizing is that even the same system at that point right once you've changed sort of the power dynamics can it be argued that we're looking uh at something fundamentally different there and i think that part of the excitement or right there's another area that we need to explore and define more terminology around in terms of that feedback loop of system system architecting system running and management and culture because right the even if nothing else right i mean i don't know how nothing would change in the system going from a centralized decentralized decision-making has to open up like there are literal operational things that need to go into account to be able to manage more people being involved in certain decision-making capacities but at the same time it inherently requires culture shift right and that's where it comes down to that behavioral level change for me because if it goes from hey daniel is our benevolent dictator in this community and no one questions it all of a sudden both of us are making choices and so are 50 other people i now need to be willing and ready to be called out for the appropriate level of responsibility that i hold and like hey why'd you do that that was actually not in line with everything else you've said before and your views on the system like why are you going like now i'm accountable now i'm responsible to my peers and i just think that i get worried about not enough of that behavioral and human level coverage and again one of the things i'm excited about with the current experimentation endows um right when i first got in in 2016 the amount of people uh myself included who were like oh dallas brand new shiny cool thing and right this is back in 2016 when there was zero tooling or anything for it and i right there were people who were there much well before then even but

there was this view it's a brand new thing and then all of a sudden folks and i know like my colleague richard brown at skirf when he joined maker dao he was in one of these positions where it's like y'all have heard of co-ops before right this isn't that like the tech might be new but the concept isn't you've heard of a co-op right and people would be like and like they had never really looked into that background and full history and so right first there's this just catch-up of right different domains have relative knowledge but then once it's these different disparate domains are aligned and this is where i really you you you call that the medigov community that we're both in right i really love the medigov community because it's not just a web3 governance community it's a governance community which means people from inherently different backgrounds with inherently different views and and different desires of the kind of applications they want to bring this back to are coming together to talk about a topic and i think once you have those layers of true interdisciplinary coordinated supported action towards trying to change a system at least that's where i get really excited about people coming together because i i do just love that those kind of different backgrounds kind of being able to chime in on on some kind of thinking a lot of places we could go as always these org streams at the academ lab it's a it's a multi street juncture there's people who might be familiar with or curious about active inference but then some of these questions and approaches they're hearing about for the first time and then there might be somebody who's interested in the governance approaches who might be hearing about active inference for the first time so that's kind of the fun it's the microcosm of how we make decisions and how the differences and what our experiences and perspectives are contribute to something that's bigger than us and i think that is related to your point about uncertainty and about changes in certainty of different types which i think we can take some time and think about um like if the plan is big enough for one person to feel like they understand it there's a limit to the complexity of that plan if you allow for stigmary for the ability of the plan to be understood by one person plus their environment like blueprints or the computer system it's bigger than that person's one on-board cognition so that's what a lot of cognitive scientists call like extended or inactive cognitive processes that can involve physical parts or modification of the niche so that's been one trend in neuroscience since the sort of like um brain region a connected to brain region b still very cool and interesting but one of the important recent trends is people looking into this embodied and extended element little aside so you have the project that that one person can operate by themselves but we're talking about projects that are either requiring multiple expertises like almost any scientific project or the case where we not just could use multiple

people we might benefit from but we're talking about governance which is like decision making where it is explicitly about creating a layer multiple people are harmonizing their view so i think where does uncertainty fit into that is a fascinating area i'd love to hear more on because uncertainty is such a core component to active inference and the management of uncertainty so maybe you could give some thoughts on that and then we can bring it back to active yeah so i think there's a couple of dimensions there that i don't want to conflate either so one thing is that right i i know uh personality psychology i i feel like let's just go with the big five to avoid let's go with the one with the most research behind it um right research uh personality psych its own big school a lot of differing opinions uh that's not necessarily the focus of the current the reason why i'm bringing it up right but there is reason to believe there is some biological difference between people and again not trying to dive into all of the nuance of that at the moment but one element is the ease of the ability to deal with something like uncertainty i do not think is biologically uniform right i think there's going to be people who have easier time dealing with uncertainty and people who have harder time dealing with uncertainty and that's just the natural disposition right then there's the actual lived environment that they are raised in grow up in the culture they're in all of that which also heavily affects that so a question that i remember seth frey who's uh professor seth frey who's also in the medigov community asked during a a podcast we had or a conference uh panel we had organized last summer uh he brought up this idea that he likes to ask people of hey you when you envision your version of utopia right imagine and i'm butchering his exact example uh but it was sort of oh yeah for dao it's sort of the the the perfect out like and envision it as a building right do you see that building full of people or lacking people and i think the answer to that starts with a big dichotomy of viewpoints around the question of uncertainty do you view uncertainty as an inherent human flaw that we have to get rid of the people and create systems that can live in spite of humans or do you embrace the fact that uncertainty is just part of who we are and we will never ever ever get rid of it fully and we just kind of have to be okay with letting the current wash over us and take us where it takes us right and i am fundamentally of the camp of like this is all quarterly human i don't think we can auto automate away human problems so i i just i'm giving my own bias here i'm very much taking this in the camp of like well let's just dig into the assumption that these are core human issues and we will have to deal with the human to human side so that that is an important distinction i want to make because i do think some people are naturally predisposed or were uh or now are predisposed due to culture and environment to be like oh humans are the bane of existence and

uh what was the jean-paul sartre quote um hell is other people right like there's people who genuinely just have a harder time for a variety of reasons being with humans which adds complexity to uncertainty right so i i don't want to pretend that everything else i'm going to say is a blanket statement for others right i myself am a bit of an ambivert i have extraversion i have introversion but i i am fundamentally human in the sense of like i need other humans around me and i will always prefer a messier conversation than no conversation so that is my bias and predisposition and to into answering your question um so on the actual side of uncertainty right so existence is inherently uncertain our parts of us want to create certainty for a variety of reasons right biologically we want uh certainty so that our brain can be less freaking out about where whether or not we're gonna die soon right on an evolutionary scale that's oversimplifying where like fear responses and whatnot came from naturally right but if you think of it on a socio-cultural perspective those did not naturally evolve over millennia or over millions of years right those were instituted and provided for a variety of very different perspectives right uh why we may have been trained that a good worker is someone who shows up to work at 9am and leaves at 5pm or whatever the norms are around whatever you're raised in right there are very different uh pulls and incentives around why that was kind of indoctrinated into your brain aside from oh this is just how the natural world works right so that's also an important gap to recognize because why i love the lens of existential psychology and psychotherapy is that there's more and more research pointing in the direction that once something is a core enough assumption for someone attacking that core assumption feels like you're attacking them as a human being and like it feels like it's a fundamental survival environment for them right you might have been in a conversation with someone where you ask an opinion where you just really disagree and all of a sudden they get into like hyper defensive mode and it's like whoa i was just exploring an idea why are you like i'm not trying to cut your head off like what's happening here and but that's how we respond when people can question certain aspects that are that are so based to us so i think that again and i realize i'm just kind of dancing around the exact question but just uncertainty is such a hyper complicated topic because on the one hand it's everywhere and it's fundamentally part of existence right this is one thing where it's interesting how psychedelics and buddhism and meditation and all these other dimensions come back to the same thing of oh yeah life is super complicated and i'm just never gonna get all of it and like that's cool and you to learn how to be okay with that right and like that's uh on the very behavioral personal level learning how to deal with uncertainty is not easy for people i know i have some folks who i absolutely adore in

my life why i tell them all the time stop being a control freak you're making your life harder right but that's not easy feedback to give someone and it's kind of useless feedback unless you actually sit down and walk them through step by step how to do it and what are the things and when are you going to be uncomfortable and when you're going to be comfortable so part of the tricky part of uncertainty is that it is inherently uncomfortable and most of us don't want to be more uncomfortable so how do you learn how to of knowledge of where things are going like how do you give up control how are you okay with knowing i have no clue where my next step is going and yet you're like oh no but this is awesome and amazing and like that's actually that is a series of contradictions internally that is hard to consolidate unless you have the disposition the privilege and a few other factors in your life to allow you to explore it and like i recognize that like i've spent the amount of time digging into these things in the last 10 to 15 years because of the specific dynamics that i was born into and other factors that like i don't want to turn this into a personal uh outward facing therapy session but i'll be like i'm happy to disclose those because i think that's the dimensions of connection right it does run that deep a lot to say there i think i'll just sort of retouch on a few points because they're great and bear reflecting upon and also because they're some of the core nodes in the active inference contextualizing and implications so existence is uncertain you said biologically we want certainty relative to our generative model to maintain persistence so this ties together like a few key things first the desire for certainty arising of that desire i added relative degenerative model which is an active inference term like a generative model just like in statistics it's a reduced representation of a complex process that spits out numbers so it's like a generative model of the weather and then you have one model that takes the data and updates the generative model and then you have a generative model that can create expected data and then compare incoming results so that's a key part of the predictive brain bayesian brain active inference and then to maintain persistence is where it ties in with identity and with thermodynamic survival so that's kind of on also on that more technical side that's what ties us to like non-equilibrium thermodynamics now for cultural entities oh if we were only a thermobacterium that uncertainty is tied up with other people yes we also are dealing with uncertainty about other things too and different people in different situations different phenomena they deal with but across the board it's pretty much other people and so that brings up some active themes like having a shared generative model reflected by a shared narrative so like we don't all have to agree what role in the movie we are but if it's different genre that doesn't even mean it's bad it just means that some people might get surprised if that's what

they expect and then some other active things we talk about are like ostensive cues which are used to signal attention and then the way that we talk about how and where attention is directed that's the regime of attention and so when the music stops after the middle of the orchestra but everybody knows that's not the time to clap what does that say about the culture and about what everyone's attention is and that's how they signal that they're all part of the same listening community for example when they do or don't take the action of clapping so social entities again whether it's just social projects like exchanging value for shoes or whether we're talking about the social media which is the governance and decision making implicitly and explicitly that's where it's such a great topic eugene and you've brought because it is about that individual micro level decision making and regime of attention i'm scanning a proposal in my web 3 you know dashboard but my literal pupil gazing and my blood sugar is going to be mattering and so how can we really think holistically about the regime of attention and the cues and the affordances and all of that for humans who we're designing for we don't want to be in the matrix pods we want them to have a thriving life which is going to be different for different people and so it's just such a great nexus that you've brought together so any thoughts and then a few more active points oh yeah i know i love that because that directly also ties into and i love the way you articulated that so thank you for running through that and i also uh an element that i that i've been thinking about in the last couple of days that feeds nicely into that uh in terms of right how do our how do our models get updated in terms of uh some of these social dynamics and the importance of exposure as a step to better managing uncertainty right i know as someone who like i'm not a a uh coder that was not what my parents showed me the the process that i went down as a kid i was not the kind to just figure it out on my own so i still have this weird fear of technology like the first time i need to go do something deeply technical i'm like i'm just gonna break everything it's like it's illogical i know it's illogical i've done enough to know that it does not make sense and each time i do it it feels a little better and sort of my internal models get updated by all the uh you know by deciding to direct my attention and understanding the fact that because this is something my existing models don't know how to accurately predict i'm going to experience a certain amount of discomfort there and taking the meta-level attention to realize that that's happening is already a great step to being able to work through more and more of that kind of uncertainty and discomfort because for me it's whatever the tools are for individuals that are important to like hey are you noticing you're uncomfortable now great next time try to like catch it a little earlier and a little earlier and a little earlier and eventually you'll hopefully start gaining enough of a sense of what

series of actions or thoughts leads to an unpleasant state and then being able to modulate certain elements of that right so again there's also the other dimension of this a privilege right i've kind of alluded to it a few times but i never want to be critical about these things because we're all coming from vastly different areas of our lives and i know in times in my life where i was you know living paycheck to paycheck and barely making ends meet it's not possible to focus on these higher order things like i i was i was very much struggling with that at those points in time now that i have a little more luxury and comfort i'm able to you know spend my free time pondering in a different way which i think is also very important to recognize when some of us think about you know financial distribution and paying it forward and sharing in abundance and all those kind of or effective altruism or all these other things at the end of the day we all have so many options to just inject a little bit of positivity for someone else and the more that we choose to do that the more we can potentially free them up to have a little more room in their cycle in a day to day to put their attention somewhere where it might not have been going before and of course it's it's utopic to think of like oh i see someone struggling i do a nice thing for them great they ponder the philosophy of life as a right like that's a lovely ideal that probably happens almost never but like more realistically right if we as a because we are inherently social because coordination is so at the core of what we do and because there's a lot of systems and other pressures around us telling us who we supposedly are right and us trying that having a navigate of well supposedly i'm selfish or i'm collective right depending on which part of the world you're coming from that could be a core narrative you hear or all these other dimensions and all of this just comes back to in my mind like uncertainty equals complexity and control in different areas right because life is wild like i don't even pretend to be able for my brain to be able to grasp the level of complexity and to be able to put it into words because it's just beyond what i am capable of comprehending but i'm okay with that and the being okay with that and the desire of control of complexity is a much more dangerous thing and directly related to that it might seem like a tangent but is the idea of like fomo or fear of missing out right it's this control desire of wanting to like oh i know these things and i have to be able to control my schedule so i can get all these supposed desires out of it right and the whole idea of knowing yourself and how to really know yourself is a whole separate thread i'm not going to go down the moment but i recently heard the term jomo and not just to give more whatever terms but like the whole idea of joy of missing out right to know that hey if i'm choosing to be here with you right now doing this sure in the back of my mind i can cycle through all the things i chose not to do as a result of this and i'm missing out on in this window

or i could just be like no i'm super freaking lucky that you asked me to be here with you and we now get to share this experience and whoever hears it gets to hear it and boom i've focused all my attention i don't have to worry about adding all those other things which in turn right the whole idea of fear of missing out and being always worried and wanting to like control these other factors where it just leads to this place where you are stealing your own attention from yourself away from the things that on your deathbed you might be realized were the most important to you and this is also to take another sidestep why in my opinion meditating on death is very important because that's one of the few things that can help you realize of like oh this was literally the last second how do i feel about everything that got here like that's one of the only ways in my opinion you can like deeply understand how truly satisfied are you with the suite of activities you choose to fill your time because a lot of us have we are shackled by things we have to deal with and that is a different set of thinking and emotional judgment that should go to that direction if you are hyper privileged and not happy with how you're spending your time you deserve a very different conversation than someone who you know is barely making ends meet putting food on the table for a family of three or whatever right like we also have to be realistic with ourselves and know how harsh we should or should not be with ourselves so i'm happy to go into more like some of my own personal psychological tools on that side but i don't want to just keep rambling well this is already quite a unique and different um discussion on active it's really interesting so few thoughts uncertainty is the core of decentralization that's what you had brought up earlier and then you connected that to treating each other nicely so it's up to us on the edge to act infer and serve the way that we expect and prefer it signals to ourselves not the least of all what kind of person we are like are you the kind of person where when the nft is worth a hundred you give five away and are you stressed about 130 because it could be higher because it's a number so it's like if we are going to have that different system with decentralization it will be not just like um part of the picture that people are nice to each other on the edges that will be the system so either we're not in that world or we are and so expecting that somebody else is going to redistribute our centralized taxes to do that is at the very least emboldening that system and then also it signals to others what kind of people they are expecting to see if you pull 10 red and one blue out of an urn bayesian statistics works a certain way to update priors it could be very tight precision or very loose precision but at some point you're going to be like are there more of this one kind than the other kind and so that's related to statistics but also the way that our brain the bayesian brain predictive brain uses things like statistics and that includes our cognitive biases as they

say but of course tobias is to take signal out of the world so what will those signals be and then also when we provide people with material support you're bringing it back to like it again it's not just a solipsistic single trajectory optimizing strategy there's like a level of strategy where it's not like the individual must lose because the group has benefited it's like it's an ensemble and one's agency and their perception and action their perry personal space has a certain spatial and temporal scale and if one doesn't care about leaving the future a wasteland then we can talk about that in this model with their expectations and preferences maybe they have other reasons to think that way but there's a lot there too yeah you know i mean just to start at the end there i mean the thing that that i think about a lot in that sense is i wish that and i'm just taking four because i believe forbes or bloomberg whoever does the billionaires list right i i wish instead of doing the list of here's the people with the most wealth in their bank encounter that they can just claim their name is directly affiliated with that we tracked who's created the most jobs in the last year who's actually distributed the most wealth on the planet like how many people get to put food on the table that they didn't afford if it wasn't for someone else or some kind of other metrics that can be deemed socially impressive that actually reside on the the social dimension and not just thinking of us as individuals because the whole yeah i mean we can i don't want to take us more off the pathway with uh just like the you know history and wealth and what not in in the states and any questions around that but for me it does i don't know one piece of research that i really appreciate in this direction is the uh oh i'm forgetting all of their names but the the the the group in in siberia far out in russia that starting in the 70s if not the 80s did the the training of the arctic foxes to see how quickly right and arctic foxes are notoriously not friendly or pleasant uh how quickly can they get them to be fully domesticated and i believe it was within 10 generations if i'm remembering correctly but roughly at the time this is still when i mean especially in the soviet union where science was complicated for a variety of reasons especially anything that had social implications to it and not just pure math or physics um but no there was not many there were not many people at the time who truly thought yes you could domesticate a wild animal shortly because there was still much more of the you know on the human level of uh of we're just rigid creatures we are the way we are we don't change so all of a sudden that we saw that hey you know you can actually very select very clearly select for certain features that get us there right the interesting question for me becomes what were the features right what's the thing that you pick generation to generation for selective mating that actually led to a much more domesticated version of the fox not that domestication is an end goal but it is one of a

few strands of research that is trying to explore on our evolutionary scale what were the actual features that led to us being such good coordinating creatures you know up until society and modern civilization kicked in because then things got much more complicated um but there's very good reason to believe that it's niceness it's kindness right and like think about a social dynamic sure there might be the super strong macho whatever you know you know not even trying to gender it right there could be the pure by traditional like there's the the person that can squash everyone else in the system right and they think of like oh we all have to defer to that person but that seems itself to be a social construct right if you actually look at what what decent amounts of scientific literature starting to point to that it is nice and niceness and that the more we can try to create a system where that is what is rewarded and encouraged am i pretending that everyone is going to become nice no there's always going to be like there's always going to be people who don't want to play well i'm just not going to let that stop me from being nice to other people right you can either have someone be mean to you and be like well why should i bother being nice if they're not going to be nice to me or i can be like yeah people are people right some are just going to suck and that's just how it goes because when you have this much uncertainty there's no way you can like all of what's in that uncertainty right if you can find certain strands of it just be okay with the complexity of it and find comfort in knowing that i will never solve this riddle but i get to be part of just playing around and tinkering with it like that alone can be really empowering and exciting so i don't know for me i i keep trying to reframe niceness as not just like a a either a you know a thing that your parents taught you to do because that's just the thing that they want you to do whereas this you know it's thing it's actually socially weak and it's i don't know i don't buy in any of that and i as someone who enjoys it i i now enjoy genuinely just being nice to people or when people are nice to me it's nice to see that there is research coming out to say hey hey there's reasons on a neural level why that feels good but also be on a longer evolutionary scale why that is the feature set that we should focus on in system design to actually get us collaborating better together but that doesn't solve the fact that there's a lot of incentives around not pushing us towards niceness and for me that means if we're trying to create systems where that is a thing we want to optimize towards but that is in a larger structure where that is not a key priority right that just means we're gonna have a tremendous amount of dissonance between those two systems and if you're trying to fight that good fight you know good just in the sense of being kind to each other um it's gonna suck right there's no there's no way it cannot unless all of these systems break down and we build a new and to me that's a much scarier option is

anything that involves like truly bringing down mass global systems i don't see how that happens without very intense international affairs that i would personally be okay without seeing so i i think that you just have to be unless you truly want world war iii and you know near extinction and then we rebuild from there i think the only alternative is we just got to suffer through the fact that there's it's going to take us a while but we'll get there eventually but just be good to people and a lot of amazing things can happen from that wgmi well thanks for bringing that important perspective sometimes it's easy to get caught up in how many papers are written on active inference and just the rhythm and the micro that's exactly what you're really calling towards so let me bring it back towards some active core terms and like ideas and let's see where maybe the niceness and some of those ideas come back into play so um we already talked about uncertainty being one of the key factors and that not being like a hot take or contrarian opinion it's like a de facto understanding that what is happening from a sensory perspective is not mere signal processing and filtering and the way that a one directional neural network processes sense but actually like an active generative process that's compared to incoming scarce data that's why we like don't have a blind spot in our eyes and things like that so that is not contentious it is a core piece of active inference however now in active inference there's action amidst uncertainty and so that's a lot like the action perception loop or the ooda loop for those who are familiar with it every single cycle every crank that's run inference and action are co-occurring and so there's a few angles that first is that we never make a system where we're expecting to do something that it's not enabled to do or vice versa so even just qualitatively before the computers come out this helps think about how subunits are connected like if the forager ant is waiting for the interaction from the nurse ant but the nurse ant doesn't have that affordance there's a missing piece in the system's design and so yes it is going to be a whole other level of emergence with fine-tuning that interaction but just at the level of thinking about what the entities and their connections are and how they influence the niche active reminds us that action and perception are integrated at that level also not unique to active inference very similar to any kind of cybernetic or agent-based modeling framework but what is different is the way that we can talk about different types of uncertainty and even things like multi-scale uncertainty or for example uncertainty about events and there's other talks i hope people go down the rabbit hole 101 ways because we've talked a lot about different uncertainty formalisms on active lab and i hope there's a lot more develop on those areas but just to say that again one could imagine like using that earlier example of being in a movie in a genre you could have high precision on the genre but then maximum

uncertainty about what is going to occur next in that like a thriller with a stereotypical enculture set of music and lights and all of that or you could know exactly what was happening oh it's just counting down the numbers one after another with high precision but not understand the narrative component so i think to connect it to web whatever in systems design what is it that we want to reduce our uncertainty on and how can we really bring our attention to the types of uncertainty that matter for humans affect for how we feel in our phenomenology and like the system uncertainty notes like if the block arrival time starts getting too wild we don't want that to go off the rail that's like a technical use of uncertainty minimization and control but then this embodied perspective on humans as active agents brings the human level of uncertainty in not everybody wants to be on a live stream not everybody even wants to have a say on a decision-making proposal how could we reduce their uncertainty about what they do expect and prefer and customize that and i think that's cognitive frameworks like active could come into play yeah i mean that's one where i guess it's also important to note that i don't know if the quote-unquote average person in web 3 and i have no clue what that even means but if if if you pulled you know there's going to be east denver next month i would actually this would be a fun fun study to run if i had the time and energy but you asked 10-ish thousand people or whatever certain amount of people are flocking to denver for this conference uh what what term is at the core of web three and i actually wonder how many of them would put uncertainty and how many people would tell me i'm a madman for thinking that that that's a core of decentralization because i think that a lot of what i hear at least is the idea of oh we're adding an automated trust layer right so that's actually minimizing uncertainty what are you talking about we're adding uncertainty we're trying to uh you don't have to trust in people you can trust in systems you know you don't have to trust in a banker at someone at j.p morgan you can trust the protocol instead and you know it's all great um but i mean i think that right that goes back to some of the other aspects we were talking about well on what level and what part of the system because if you're actually talking on technical architecture uh yes you could just have less humans needing to interact at a certain point but if then you're talking the complete set of full human interactions using that protocol it right this a few years ago there was the debate on what's more relevant for blockchains off-chain governance or on-chain governance you know like can you auto automate things that it's a quick yeah or an and it's all automatable and it's all interacted uh purely through code or do you need humans talking to each other and i feel like there are some you know like interest rate going up or down yeah sure those things might be on chain

governance oriented anything that actually involves more complex questions or even more complicated right with the distinction being that complicated is still modelable complex has emergent behavior um right even with complicated things that could already kind of get get challenging uh in terms of just getting humans to agree on things so um i i am not a big proponent of uh the goal of all of this being humans have to decide less i think that yes it's always great to you know do we right there's a few aspects here which i'm myself uncertain about right there's an in there's almost a unstoppable force that is more technology advancements of technology automation to a certain degree is happening to the world to us whether or not we like it um the question is what are we trying to automate and i feel like that's still very much an open question and i i am i don't know i the hmm let me see the best way to move this forward i think at its core i've heard one thing the idea of its coordination all the way down right in the crypto space or web 3 space whatever that at the end of the day what is the most base thing that huma that as a species we are doing we are coordinating right because if we're already taking forgiven that we exist uh we there's many of us here and we have desires then we have to ask of like oh well how did we get here right it seems that coordination was a very key part of differentiating us uh from some of our other ancestors down the line and so for me coordinating is inherently uncertain right just people being people together is an inherently uncertain task because unless you heavily constrain and force people to be less themselves right if you're telling someone don't be yourself be this thing i want you to be then then you're minimizing uncertainty right you're telling like i don't want daniel i want a version of daniel that i would like to see right that is minimizing uncertainty but right the whole idea of seeing people for who they are and where they're at right that that's adding complexity that's adding uncertainty that's saying i don't need to know where you're at i just need to recognize the fact that i will never know where you're at because i'm a different human i have a different lived experience than you all i can do is with my limited abilities try to project and understand what's going on with you and use a mix of empathy right can i just feel those emotions and compassion can i rationalize that feeling and do something about it and for me compassion is much more important than empathy because empathy is deep and psychologically taxing and very limiting in terms of the amount of people we can direct it towards compassion and i just finished uh rutger bregman's uh humankind book right where he and it's not as though it's his idea but the whole idea of rational compassion being much more of a scalable concept than pure empathy because if you try to empathize with all people on the planet that's just not going to work in any way possible right rational empathy can be

something that we utilize as a more focused rational compassion excuse me it can be more of a tool in our tool belt to not just fall into tribal camps but to try to actually think of how can i see your pain as a person who is totally different than me in every way shape or form has our pain and that kind of reframing is a very very powerful thing but i also want to i want to jump back to what you started with because i i know it's a active uh it's it's a given in certain circles but i for me that's actually part of the problem is where you were talking about you know the whole idea that like we don't know our realities or brains or projections or uh prediction machines all these um you know like you mentioned the blind spot like i don't know do most people realize that we literally have a blind spot in our vision and the only reason we still think we always see things is because our brains predict over it effectively and i wonder if you came up to a person on the street when you're like hey look around that's not objective reality would they tell you that you're on drugs and crazy or would they be like yes yes let's have a scientific dialogue right and i excuse me for the the previous statement but like it's the point being there right that people don't react kindly at least i've had that reaction a lot of the time even with well-educated people so it's not a pure education perspective and for me getting people like that comes back to people embracing uncertainty right how do we encourage people to recognize that i fundamentally don't know anything about anything hopefully through education and a lot of intention about my attention i will try to learn a few things but in the grand scheme of collective knowledge i will never know anything and like that's a weird thing to admit to myself like as much as i really try hard i can hopefully learn some concepts and frameworks that broadly apply to whatever environments i find myself in but in the grand scheme of information even if you have photographic memory you practically know nothing just because there's so so much of it and i don't know for me that's also been more of a liberating thought whereas when i was younger it was always a very fear-based like oh no that just means work harder whereas now it's like no just embrace the uncertainty embrace the washing over of i will never do everything so just be intentional about your attention and what you do want to do and make the most out of that so i feel like i went on a hard tangent on that one but that just felt important to call out oh do i have some some points directly to this so thank you for sharing it so yes there are niches on the internet or in real life where it is um a beginning point that the brain is a predictive entity or at least could be modeled like that which we discuss is the difference between instrumentalism are we just applying statistical models and we're just making no claims to the territory we're just talking about maps we're like cartographers are we making ontological statements like the brain does active inference

realism instrumentalism is like i find it useful to study the brain using this model so if someone used a linear regression they wouldn't say the brain was a linear regressor we're doing that with instrumentalism for active also there's realism so that's like the one point to make but you pointed like how some people resist that um sense of re-aligning their generative model to this idea that we are experiencing our generative model so i think there's a few pieces the meta is um we've all been wrong before we've been convinced on something and been wrong so your emotional conscious experience which is only the tiniest fraction of the iceberg just simply can be incorrect by virtue of the way you've had interactions in the past um so it doesn't need to be framed as right or wrong there are some useful components to thinking about the brains of predictive machine and it gives us unique predictions and explanations so for people who enjoy those features this is a useful way to think about the brain so one example is like vision where the anatomy of the eye is clear that there's a blind spot that's off on the side and also there's different resolution and color perception across the eye so we're perceiving with our mind not our eye but of course input to the retina and light activation matters it just turns out that there's a lot more of an actively updating process and it turns out that's the one that we're perceiving without the blind spot rather than the incoming sensory data and that also applies to our other sensory features what's cool and closes the loop between individual decision making and behavior and like science as a human endeavor is that it turns out that in order to get good information about that generative model one has to do informed sampling like our eyes are circading around we don't perceive that so that's another example of why we're perceiving the stable version of vision which is not being received by any sensor and it circades around the vision to reduce uncertainty goes where there is visual uncertainty and so that generative model isn't just like a sort of one frame per second of what is predicted to be there it also carries like extra information like where are you uncertain in your visual field so that is like the process of being a scientist where you're getting experimental observations but then there's a system you didn't observe sometimes you destroy the cell to measure it or you have to kill the animal or whatever it is but there's observations and then there's the hidden system state and it turns out that that maps onto a lot of statistical models like the partially observable markov decision processes which integrate that separation between observation and hidden state inference in the context of action selection so that's kind of the family that active inference is in and then just the second point was about the empathy complex online systems and another active inference term is this thinking through other minds so it's related to theory of mind just like it sounds and it's referring to when one's uh

counterfactuals or their cognition is based upon how they expect something will be evaluated by others i will do x with my clothing or speech this is it could be implicit or an explicit uh self-talk because others will see me this way because my partner will see me this way or because my dao will see me this way so that helps us get a handle on complex social cognition if we can have models of cognitive entities that can engage in reflection and have anxiety and then um that's where roles in complex systems might come into play so you said that roles reduce i agree sometimes they reduce uncertainty about one node but then they expand it like you're gonna be the saxophone soloist and i'm looking forward to how you improvise so it's like we reduce the group's uncertainty who's gonna be the soloist but then that's what opens up the certainty um distribution in another way or like we're playing chess when you can have brilliant games of chess because there's constraint and there's like roles and but if if people were always mutating the games of chess there wouldn't be as many classic games because like the structure would just be a lot more superficial and so roles might be able to help us reduce uncertainty about certain features and then humanize because oh they're a learner in this meeting amazing citizen science is so great i love working with people who are just learning about research and then everyone can feel included in that even if it takes them 15 years to learn linear algebra in that one meeting it could be very okay and nobody would be getting graded for example and i think that especially the last uh element that you were talking about there i think it's important to stress the difference between embracing uncertainty and maximizing for uncertainty because i feel as though people might unintentionally hear the idea of embracing uncertainty as as that is an implication that more uncertainty is always a good thing um where i actually think when you're designing a system if you want people to flourish giving them too much uncertainty is a debilitating factor so right there's the there's the system design component from the actual function of the system and then there's the cultural psychological component around it and right on the cultural psychological component that's where as a designer of a system or community or company or whatever you think of how you set that cultural tone and you set right you by interacting with people in in various ways that you do in your organization you set a certain tone of are you embracing uncertainty or do you always want to especially if you're the leader in your organization do you need to show this facade of like no i'm always certain on everything right because that actually sets a cultural tone of it's okay to be fakely certain right like there's no way you're certain about everything i think every leader is inherently uncertain about things because just of how many options there are but you also need to know when to be okay with that uncertainty and in spite of it to still

be like no we're doing this one thing and that's the direction we're headed it right like that's its own its own interesting aspect but i i think when it comes to designing any system whether it's web3 crypto whatever you're you know you're carving out an internship for someone like whatever it is but creating constraints and a sandbox is very important for people's comfort level so i think the way i like to think of it is the amount of constraints need to be greater while people are newer to things and the more they're exposed to it potentially the more you can open up for them right again not everyone is going to be not everyone thrives in having uncertainty some people need their hand held through it and then they can start thriving some people at least i i don't have enough evidence over a long enough time period to know but there are some people for whom adding uncertainty just everything collapses and crumbles around them and like their role in the system just cannot proceed and like if you're building right you can think of dows as these fantastic new futuristic these are employment opportunities for people right like people need to buy food and they get it now from decentralized decentralized ecosystems in some capacity so right you can't this is where i actually i still am unclear on how i think because if you go to a lot of these communities and you're like oh how do i get started just do stuff and that's great for the privilege right what happens when someone's like well i hate my job i think i can help you here i want to help out but i need you to like help me help you out because how else am i going to pay for my kids food tomorrow and i can't make a change in my ecosystem and i can't just spend right i don't know how many hours your your listeners have spent trying to get onboarded into various dows it can be a painful experience right a lot of these groups were started by a bunch of brilliant people who come together and it's other brilliant people who get pulled in and then all of a sudden when it's well what's the ramp way for someone who has no clue what the heck is going on here that was an afterthought and that was kind of produced after the fact when someone complained about it or once we had enough time and so yeah uncertainty is this fascinating concept because you have to know when adding more of it is a good thing versus when adding it is a bad thing and you have to know from a system design perspective where do you culturally embrace it versus where do you actively try to get rid of it at all costs because that's what's going to help your team thrive and you can still embrace it and want to get rid of it in certain areas like those are not inherently contradictory right you could be like i don't know what the next year holds from the market but in this specific function of operations where i'm going to hyper control this in every possible way so that i know that outcome with high probability right think of anyone in a supply chain if they tell you where they know

what's happening with the resources three years from now they're lying no one knows what's happening three years from now right but they could model it better than someone else right but in their actual supply chain they could control that fabricated room so that they literally know every speck of dust in the air right and that that might be worth mitigating uncertainty there while still embracing it in the longer context okay few points first in active inference where we put action on an important grounding with other kinds of inference processes it's not just uncertainty about the outside world which of course it's easy to spiral on it's actually also uncertainty about our action so huddle i have high precision on my action policy with respect to btc no matter what color on that site that day so that helps uncouple and put space between or it's staked it's locked or it's a multi-sig so that can offload individual level uncertainty and anxiety onto programmatic systems so then the certainty in systems design so i don't remember who initially told me this anecdote but like the people who we today call politicians will say things like we're not going to sleep until this happens and we're gonna it will happen but no one says it's so important the bridge be designed properly that we're not gonna sleep and it's a serious issue yes lives are on the line we're building a bridge so how do we do it we have bridge management and design and processes around making sure that it actually works so as governance becomes less about the virtue signal of we're not going to sleep until we figure this out and more about we're actually doing systems engineering we can treat it more like the bridge and say yeah oh yeah we're all here we direct a regime because we care about the people on the bridge and that's why we're doing it this way and so that will be a huge difference than just who can signal loudest and then just one other point was like research to bring this back to decentralized science and research is a multi-year task like i've been doing entomology insect biology and insect genetics for 10 years and i'm only beginning in this area it's cool to be excited about many areas that's why there's always been like complexity and it's awesome to be open-ended and love exploring those areas but for research and application like the cutting edge it is a multi-year upskilling reskilling process if you're not on the cutting edge and on the cutting edge it's just continually that plus sometimes even more so being okay with all of that and then just recognizing that it's a multi-year process and whatever mechanisms for whomever and when and however we need something that's going to not just rapid assemble the bounty team and then do the multi-sig and then leave some people might find those opportunities appealing but we need to recognize that a huge amount of this is going to be like multi-year communities and the kinds of relationships and small teams that will only scale if we're all engaged in it yeah yeah that makes me particularly think about two aspects kind of

community size and temporality of change or the time dimensions of change and on the on the community size front and again right this is i this is not an area where i think anyone has the answer right this is just me me kind of uh giving my own views and personal biases here but i i strongly believe in especially just exploring what i've been exploring in this world and generally um trying to learn in the direction that i've been trying to learn what i get excited about dows is actually providing the digital infrastructure for a super nuanced global landscape of communities of communities where at the end when it's humans actually organizing and making decisions together actively or at least most on kind of the consumer end so to say that those are going to be in in tiny pockets and above those pockets will be bigger and bigger communities and points of connection that are more global and bring them all together right the whole idea of say for something like a supply chain how can you have a system that you have the global benefits right for consumers it would be great if amazon covered all of the global distribution right for the people working in those factories it would not be great for that and possibly for our planet i don't know on their global footprint well enough but right there is this idea of how can we get to a point where we have global efficiency around a variety of topics while still having very localized interaction and nuance and this for me on a very abstract level is why decentralization is important because we are currently barreling towards a world of more and more centralization right it's the monopolies because or whatever i don't want to have pointed terms but you know companies tend to have more market share they try like their only goal is how do we not just grow the pie but how do we make the biggest share of the pie hours right there's a lot of fundamental incentives that are not the healthiest so to say but at the same time having global efficiency around certain functions such as a supply chain such as global money such as global identity would be great but having those certain infrastructures available on the global scale is very different than what you would actually the environments you would organize in as an individual and i i don't want to be part of 50 million person communities because then i'm going to be overwhelmed and i'm going to have no clue how to navigate it and i have no clue what the heck to do in that i'm just going to find my core group of people of i don't know a few a handful to a few hundred and somewhere in there and that's probably going to be more those size communities that i play in so bringing it back to more dows and especially dow research dows right now uh so i can also share with you the link i don't remember if i did ahead of time but i'm just putting together a different organizations that are in one way or another playing around with uh the intersection of dows and science in some kind of capacity and some of these groups have

nothing to do with web3 except that they're using it as a tool as a mechanism of coordinating and funding right the actual science is all non-related to web three um and that gets me super excited and i think uh thinking from the coarsest aspects of like us as a group of scientists where do we want our research or whatever we're doing to have an impact great right and then for people like me who for whatever weird reason love building infrastructure or just like open source and just like operational layers right like how do i work with those separate pockets to be like hey you all shouldn't be worried about the operations there should be a layer of operations wrapped around all of the signs that help you do the operations and there should be a fundraising now wrapped around all the signs now that can connect to all the main science funders across the planet and right there are these mechanisms that just incentives have never existed there and i honestly don't think they're there yet these are still from my vantage point these are still public goods that in any other context but web3 would comfortably say yeah we need tax or charity to make this run but web 3 being web 3 those are both bad words right so we have to pretend that it's oh well we'll just keep having money trickle in because right now money grows on trees and no one's working about worried about sustainable funding 50 years from now right but at some point uh this fun part of our dance comes to an end and we have you know we're not just growing on on the huddle beliefs at some point at some point we recognize like no we've reached probably the zenith right and at that point how are we actually going to fund anything that's an infrastructure layer and right now there's some really interesting experimentation i know even metagov the community we've mentioned they have a doubt a fun research in the direction of governance like i love these kind of experiments but again peer review and that's one that we want to take a serious focus at at smart contract research forum because we don't see who's playing in that direction we see some people playing on the publishing side we see people wanting to do the research side but right but the core infrastructure is just it's less sexy it's not as fun to build and you don't get to have your name out on things right so like that that's where it's like i i that's where we want to take a focus on what we want to add but i want to be respectful of time because i know we're getting dangerously close to the end here let's have some last thoughts and then we should we can do a dot two we can do more so um yes bolts business operational legal technical social that's one acronym that we use those are all important for research and if the person who has the most knowledge of equations is doing the accounting it is not efficient allocation and then font like formal documents ontology narrative and tools so those are some of the frameworks first thinking about systems design and then review um well here on the live stream where we've

done like 36 paper discussions over the last two years or so we often wonder about where that could go so um having 10 people look through it and different people do different thing one person runs the code one really digs in the citations it's a live stream people are rewarded by participating in the lab we're learning by doing i'm spending more time reading active inference papers than writing them by far and most researchers probably are too otherwise one starts getting disconnected from the pack so how can we make that contribution those notes that people make in review not like accept or reject but yes and and then we have other people's perspective on something it doesn't mean that it's right so there's so many things that can be done with knowledge annotation and that sounds like a great area to look into i totally agree that there's been a lot of focus on financialization tokenization um conceptualization in research and the parts that i would like to hear more from is that recognition of the multi-year challenge and about what kind of research people want to specifically do and spearhead and hold on tight through because even when you're getting paid as a grad student in a really lucky situation it can still be challenging so when those features like health care and weekly check-ins with a mentor aren't in place that's the situation where research will be happening so um [Music] what are your last thoughts or where do you think we should go from here yeah i mean i think right careful experimentation i think is very important right now and i also uh i think it's important for the folks who are experimenting to be aware of the potentiality of their experiments and i think we've all seen the how move fast and break things without any concern for potentialities of outcomes can lead to very bad outcomes so you know for folks who are experimenting with something that realistically is almost has a you know a sub 1 probability of ever having a meaningful negative impact on a person yeah go crazy have as much fun as you can experimenting in that direction right the more it's actually closer to where people are making their livelihood and are spending their lives and that it can negatively affect a lot of people the more that review focus and the more questioning there needs to be about how do we make sure that we're doing this right so honestly i'm just really excited we're still early in 22. uh it's been a whirlwind these last few years in so many different ways you know i'm just hoping that there's more chances for collect uh coordination for collaboration and especially for folks in the in the web 3 space i don't mean to make whatever predictions at some point the market is going to turn right i'm hoping we haven't started yet maybe we're already at the beginning of the bear market who knows i'm not an i'm not going to pretend to not financial advice not scientific advice not legal no no no formal advice human to human personal advice maybe nothing beyond that right but i'm very interested to see once we hit the

bear market what does the collaboration look like because i think that's actually going to be where a lot of gold comes uh and a lot of just intellectual brilliance comes out because then a lot of the noise is going to run away so i know a lot of people because their bank accounts are horrified of when the market turns and i get it but i actually think it's going to be very good and healthy for the industry over the next three to five years and the way you raise so many important issues big thanks for that and with the review it's almost like we are focusing on the publication and the research and the funding sides but that piece of review it's almost like the first one we need that is a strange loop of science and we need review on the plans for desai and review on the frameworks for building the plans for desai and on all of that and so that's like review guild is really a strong place for people who want to learn by doing and be a big leverage point it's like the coefficient of the grants and the way that knowledge by people who are off doing field work gets incorporated is through review especially in these new structures which we probably haven't even seen yet but they're not going to look like proceedings of the uk 1600 alchemists it's going to be quite different that is for sure yeah and for anyone if you are interested in in peer review or these kind of decentralized science centralized peer reviewed decentralized publication of research please let me know always feel free to reach out and i love connecting with folks and talking to a lot of folks who are trying to solve this problem from different perspectives and trying to figure out how we all come together to at least share our learnings if nothing else and i'll definitely make sure to coordinate with you in the active community on that side let's do a dot two with a panel in a in one or a few months let's check back in on some of these themes revisit the slides hear from some new perspectives update on the projects that we talked about today and just see where we're at sounds like a plan thanks for having me again daniel thank you eugene awesome conversation much appreciated bye you